Fuelinex WDC

Last Modified: 24 February 2024

Date of study

February 2024 -- > December 2025. Analyses started in early 2026

Date of Public Archiving:

2027 (Estimated)

Goal:

Investigate how extended growing seasons influence tree's capacity to grow to following year.

Contributors

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General Files

File	Where	What
greenHouseDesign.pdf	Github	Greenhouse arrangement by blocs for Fuelinex 2024. Might change for 2025.
dailylog (folder)	Github	all .tex files necessary to run Fuelinex daily log activities. Run main.tex without moving the folders and files.
IP_adresses.md	Github	IP adresses for the climate chambers. Useful when using VNet to control them.
schedule (folder)	Github	.sch files for the CoolSpring and WarmFall treatments. This is the file extension necessary to load them in VNet.
BalsamPoplarPhenologyObservations inCommonGarden.docx	Github	Phenology guide kindly provided by Raju Soolanayakanahally.
OakPhenologyStages.docx	Github	Phenology guide kindly provided by Raju Soolanayakanahally.
budsetdataFINAL_JS.xlsx	Github	Pheanoflex budset data with pictures.
climatechamber_pots.skp	Github	Sketchup file to figure out how many pots I could fit in the chambers
photoperiod_schedule.xlsx	Github	Sunrise and sunset schedules for the chambers
sunrisesunsetvancouver.csv	Github	Sunrise and sunset data in Vancouver to have a sense of what to set the chambers at.
committeeMeeting (folder)	Github	Committee meeting notes
dailylog (folder)	Github	daily log mostly of fuelinex activities but also some other MSc stuff from early 2024 to end of 2025
whatdata.xlsx	Github	As of early 2025, what data did I have

		for fuelinex
chaptersDesign.pdf	Github	Full MSc thesis design
Fuelinex_Design2024.pdf	Github	2024 experimental design
fertilizeraddition20240607.xlsx	Github	fertilizater addition in 2024
fleetingNotes (folder)	Github	Litterature review fleeting notes
greenHouseDesign.pdf	Github	Greenhouse bloc design with hobo loggers
masterstimeline (folder)	Github	ganttchhart and stuff
a3-ruler.pdf	Github	Paper ruler pdf
biomassProtocol.pdf	Github	Biomass protocol
comitee_email.md	Github	emails to send to my future committee members
comiteemberslist.md	Github	List of potential committee members
grammarlyTest.tex	Github	My usage of grammarly as per lizzies request on AI policies
segireplication.md	Github	Segi replication as I changed the nitro boosted trees (they ended up never receiving any nitrogen boost in the fall) to non boosted ones to increase replication as so many died.
tree_ordering (folder)	Github	tree ordering
awards (folder)	Github	All the different awards and conference application I submitted during my MSc

Data

File	Where	What
		Hobo loggers that were set

hoboLight (folder). 22 files according to their position in the greenhouse. B1, B2, B3 and roof. See greenHouseDesign.pdf for the greenhouse organization. *	Github	up at totem field and in the growth chambers. Each date folder contains the data when it was logged in as a back-up.
hoboTemp (folder). 5 files recording the temperature log in the climate chambers for the CoolSpring treatments in 2024, followed by the temperature recording at Totem. See greenHouseDesign.pdf for the greenhouse organization.	Github	Hobo loggers that were set up at totem field and in the growth chambers. Each date folder contains the data when it was logged in as a back-up.
2024MiscellaneousMonitoring.csv	Github	miscellaneous monitoring spreadsheet. This is a condensed version of the different problems trees have had so far in 2024. E.g. dead apical shoot, chlorosis, light stress, mildew
2024BudburstToBudset.csv	Github	Bi-weekly, weekly monitoring of phenological events for 2024 from budburst, to leaf unfolding to budset.
2024Senescence.csv	Github	2024 senescence monitoring
TotemField_30Years.csv	Github	30 year climate data at totem field
Jun2024TreeInventory.csv	Github	inventory of leftover trees next to the greenhouse
TreeMeasurements.xlsx	Github	Diameter and height measurements
hoboLight (folder)	Github	Hobo light loggers that I installed. Probably will

		never use this data.
hoboTemp (folder)	Github	Hobo temp loggers that I installed by date of backup.
2025-212_Chltoocomparison.xlsx	Github	Minolta vs ccm200plus chlorophyll measurements at different doy on different species so we can have conversion rate
2025-235_Chltoocomparison.xlsx	Github	same as above
2025-290_Chltoocomparison.xlsx	Github	same as above
2025Leafcount.xlsx	Github	Leaf count
2025SummerChl.xlsx	Github	Number of leaves in 2025 on 3 species
2025VWC.xlsx	Github	VWC measurements
MiscellaneousMonitoring.xlsx	Github	Disease monitoring. Mildew and sawfly on the trees
tagWeight.xlsx	Github	Randomly weighted some tags. Will average and remove that weight from the above ground biomass.
2025Senescence.xlsx	Github	Empty spreadsheet. What follows are each senescence monitoring in 2025 with the corresponding DOY. E.g. (2025_340) is doy 340 in 2025
2025BudburstToBudset.xlsx	Github	Full pheno spreadsheet in 2025 where each column is the doy and the data, the numerical phenophase
		2024 spreadsheet with chl

2024Senescence.xlsx	Github	and percent in different columns
2024BudburstToBudset.xlsx	Github	2024 budburst to budset monitoring
2024ShootElongation.xlsx	Github	Shoot elongation monitoring in 2024
2025ShootElongation.xlsx	Github	Shoot elongation monitoring in 2025
TotemField_30Years.csv	Github	Totem field climate station
biomass.xlsx	Github	Above and below ground biomass monitored in the fall of 2025
prviRedline.xlsx	Github	See readme.md for details on this special spreadsheet
TreeMeasurements.xlsx	Github	All the different height and diameter measurements à

Code

Code		What
hobotemp_cc.R	Github	Cleaning data and organizing temp data from hobo loggers
hobolight_cc.R	Github	Cleaning and organizing light data from hobo loggers
phenostages.R	Github	Phenostages data visualisation from budburst to budset
shootElongation.R	Github	Shoot elongation data visualisation
temperatureFallWarming.R	Github	Code to calculate temperature of warm fall treatment based on historic temperature records at totem field.

temperatureSpringWarming.R	Github	Code to calculate temperature of cool spring treatment based on historic temperature records at totem field.
fullJointCarryoverModel.R	Github	Full treatment model with retrodictive checks and allometry model combined. This should be the main model code
empData_allometry.R	Github	Just the allometry model
carryoverModel.R	Github	Just the treatment model
tools.R	Github	Some functions I used in wildchrokie and coringtreespotters too. There is some posterior summary functions I made with claude, color palets, etc.
plotsFuelinex.R	Github	Empirical data plots
practicePlotsFuelinex.R	Github	Nolan making some plots for senescence and biomass
colPalets.R	Github	Different color palets I collected over the years
simDataFuelinex.R	Github	Simulated data for the treatment models
cheatSheet.R	Github	Some functions I like but never figure to remember
simDataFuelinex_allometry.R	Github	Simulated data for the allometry model that I never figured how to make them work
phylogeny.R	Github	Some quick phylograms
chlComparison.R	Github	Chlorophyll comparison model in rstanarm to get a conversation rate between the two tools
cleanAllFuelinex.R	Github	Pipeline to source all the data cleaning files which are below
2024cleaningPhenostages.R	Github	Cleaning 2024 phenophases
		Cleaning 2024 senescence monitoring. File

2024cleaningSenescence.R	Github	reorganization a lot messier than the 2025 organization
2024cleaningShootElongation.R	Github	Cleaning 2024 shoot elongation measurements
2025cleaningPhenostages.R	Github	Cleaning 2024 phenophases
2025cleaningShootElongation.R	Github	Cleaning 2025 shoot elongation measurements
cleaningMeasurements.R	Github	Cleaning, heigh, diameter and biomass measurements
fullModel_prob.stan	Github	Full carry over and allometry model, but with a probabilistic function of the allometry model
fullModelpos.stan	Github	Full carry over and allometry model, but with a deterministic function of the allometry model
allometryModel.stan	Github	Just the allometry model
carryoverModel.stan	Github	Just the carry over model